



Objective

Strengthen candidate generation for ontology alignment.

We study:

- Candidate recall before ranking
- Lexical enrichment (DragonAI)
- Weak semantic priors (SeMRA)
- Representation learning (SapBERT)
- Impact on BERTMap scoring

Candidate generation is a hard bottleneck.

What Is a Biomedical Ontology?

A structured vocabulary of biomedical concepts.

- Concepts
- Hierarchy
- Standardized meaning

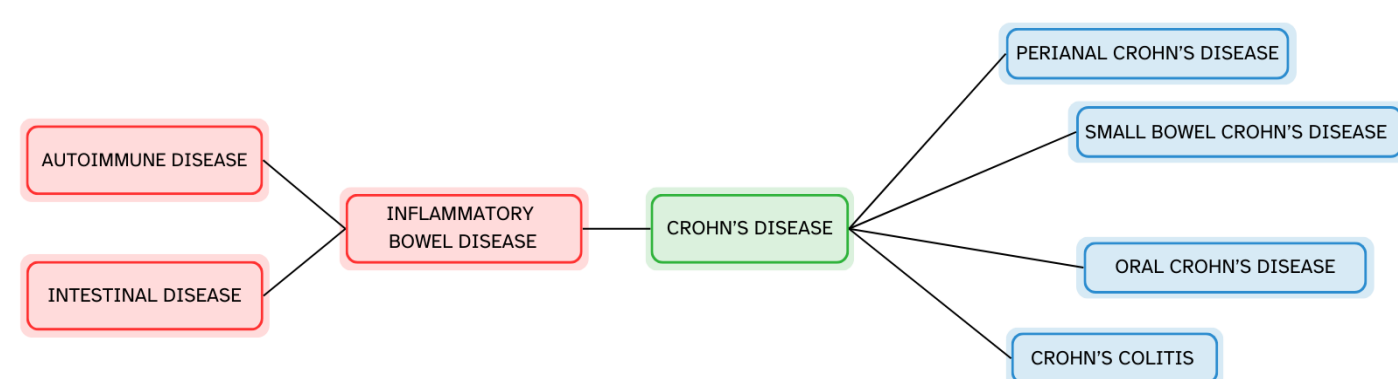


Figure 1. Ontology & Alignment

Proposed Pipeline

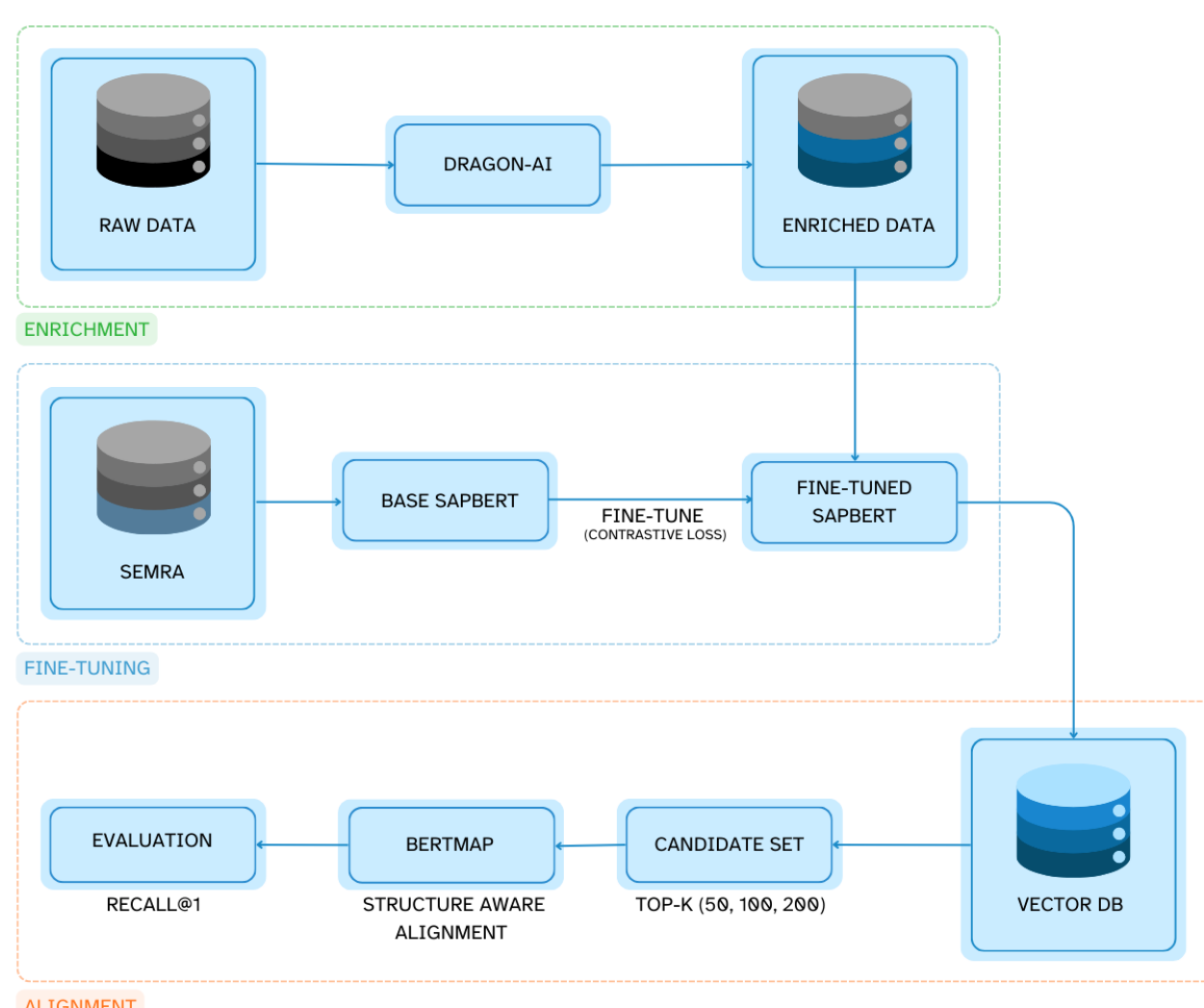


Figure 2. Candidate generation and scoring pipeline.

Pipeline Components

DragonAI Enrichment

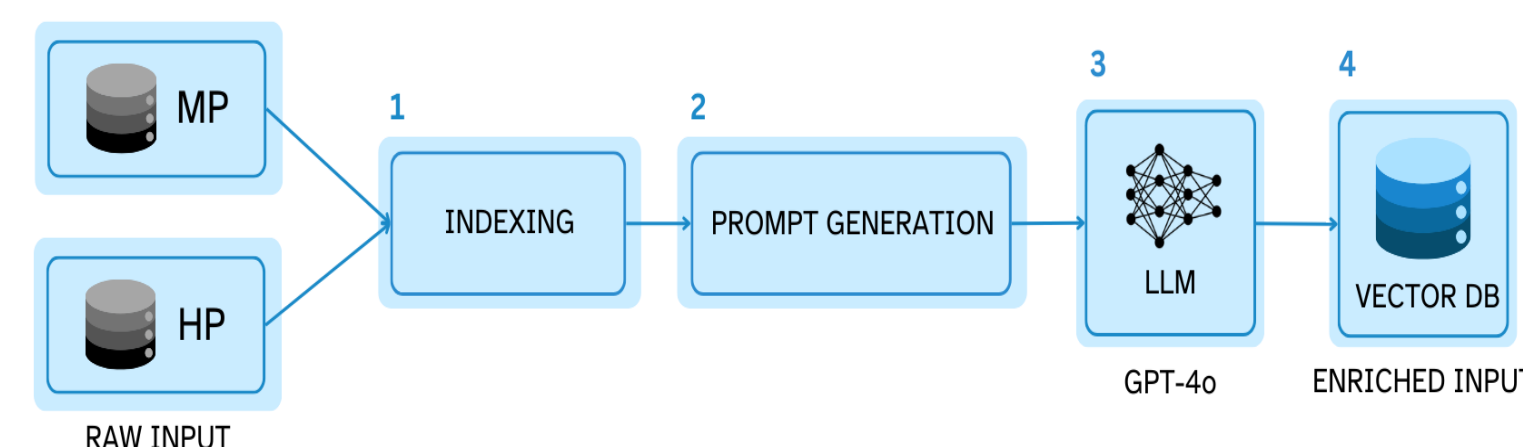


Figure 3. DragonAI Pipeline

DragonAI enriches ontology concepts with missing lexical information.

- Identifies missing labels, synonyms, and definitions
- Produces enriched text for retrieval

Enriched representation and consistency filtering

$$E(c) = \{\text{label, synonyms, definition, examples}\}$$

$$S(c, d) = \alpha \text{sim}_{\text{text}} + (1 - \alpha) \text{sim}_{\text{graph}}$$

SeMRA: Weak Semantic Prior

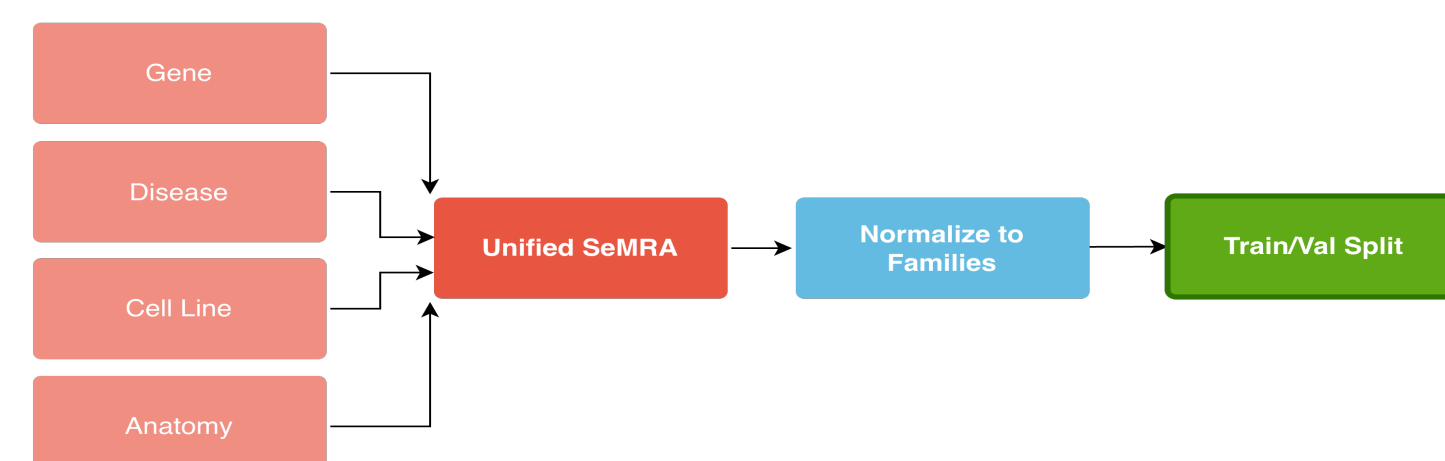


Figure 4. SeMRA pair curation

SeMRA provides high-coverage semantic links across ontologies.

Concept pairs are scored using:

- Lexical similarity of labels and synonyms
- Structural context from ontology neighborhoods

Curation:

- Retain high-scoring pairs
- Filter, deduplicate, and split
- Resulting pairs are **noisy and high-coverage**

SapBERT Fine-Tuning

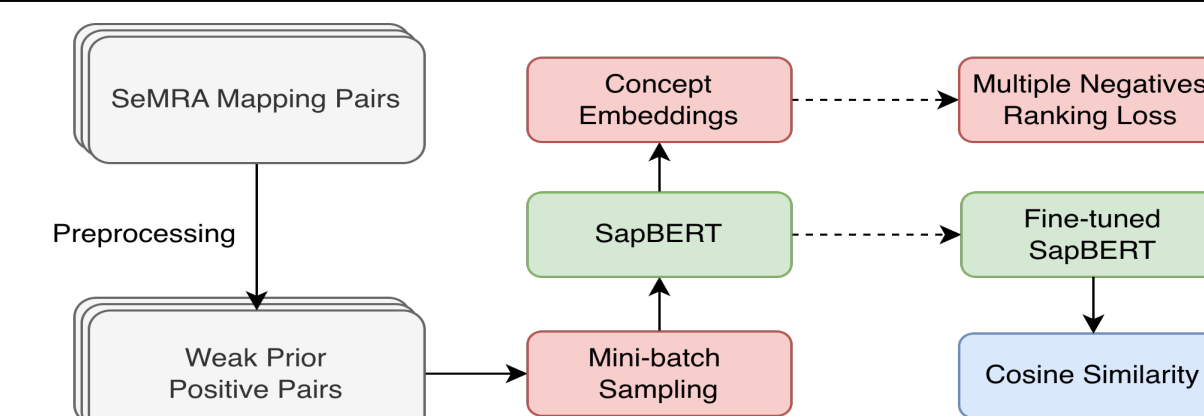


Figure 5. Fine-tuning procedure

SapBERT is fine-tuned to improve cross-ontology candidate retrieval.

- Use curated SeMRA pairs as **weak positive supervision**
- Learn embeddings optimized for mapping

Training follows a **positive-only contrastive setup**, which is robust to noisy supervision and avoids explicit negative labeling, in-batch examples acting as implicit negatives.

Contrastive Training Objective

$$h = f_{\theta}(x), \quad \mathcal{L}_i = -\log \frac{\exp(\cos(h_i, h_i^+))}{\sum_{j=1}^B \exp(\cos(h_i, h_j))}$$

Fine-tuning Result

Cosine Similarity

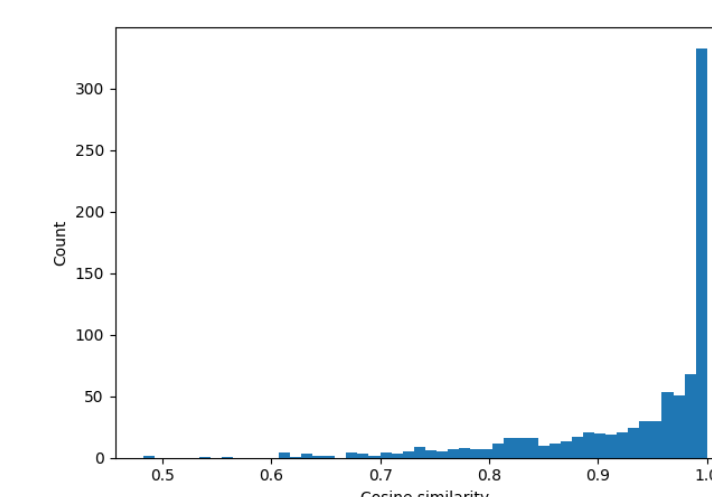


Figure 6. Base SapBERT

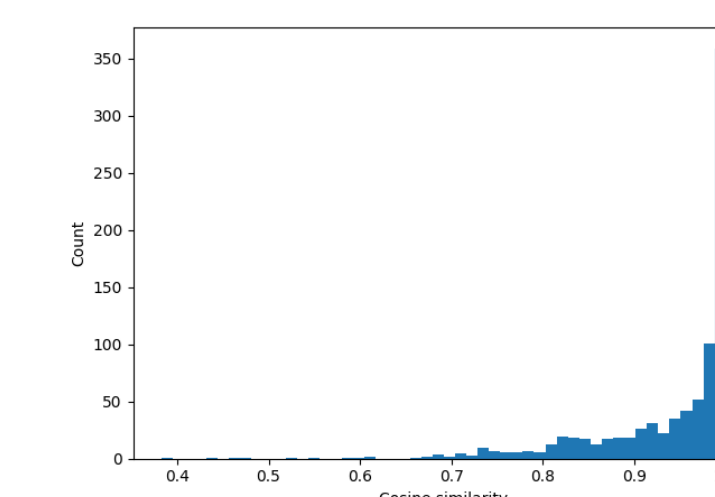


Figure 7. Fine-tuned SapBERT

Representation space

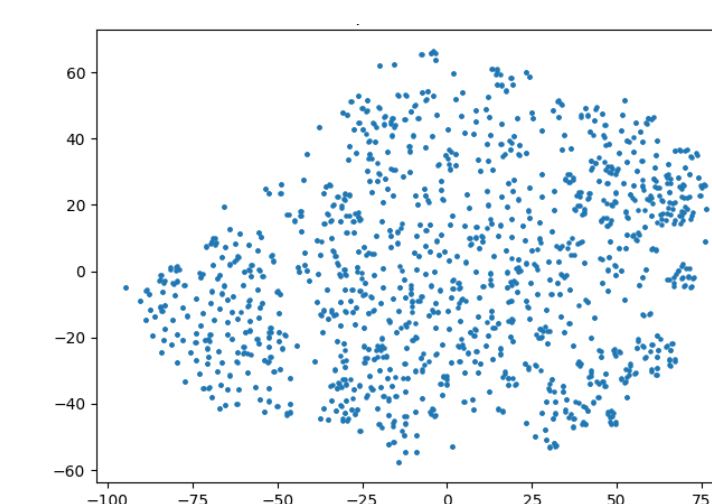


Figure 8. Base SapBERT

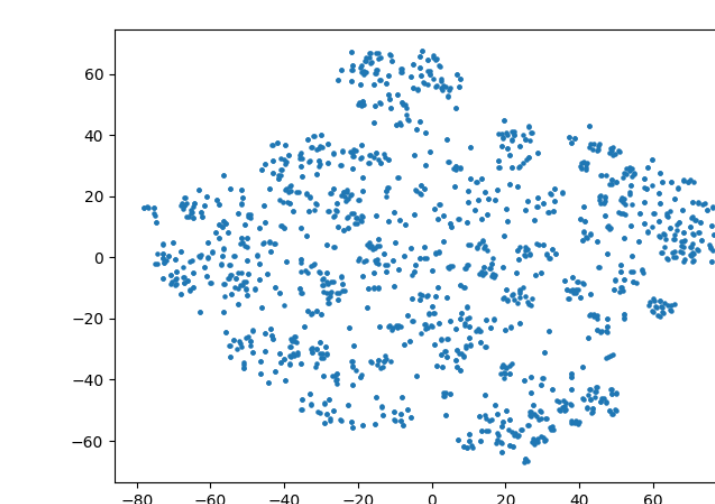


Figure 9. Fine-tuned SapBERT

Candidate Generation & Evaluation

Task: Retrieve ranked candidates and measure whether the gold HP-MP match appears in the top- K

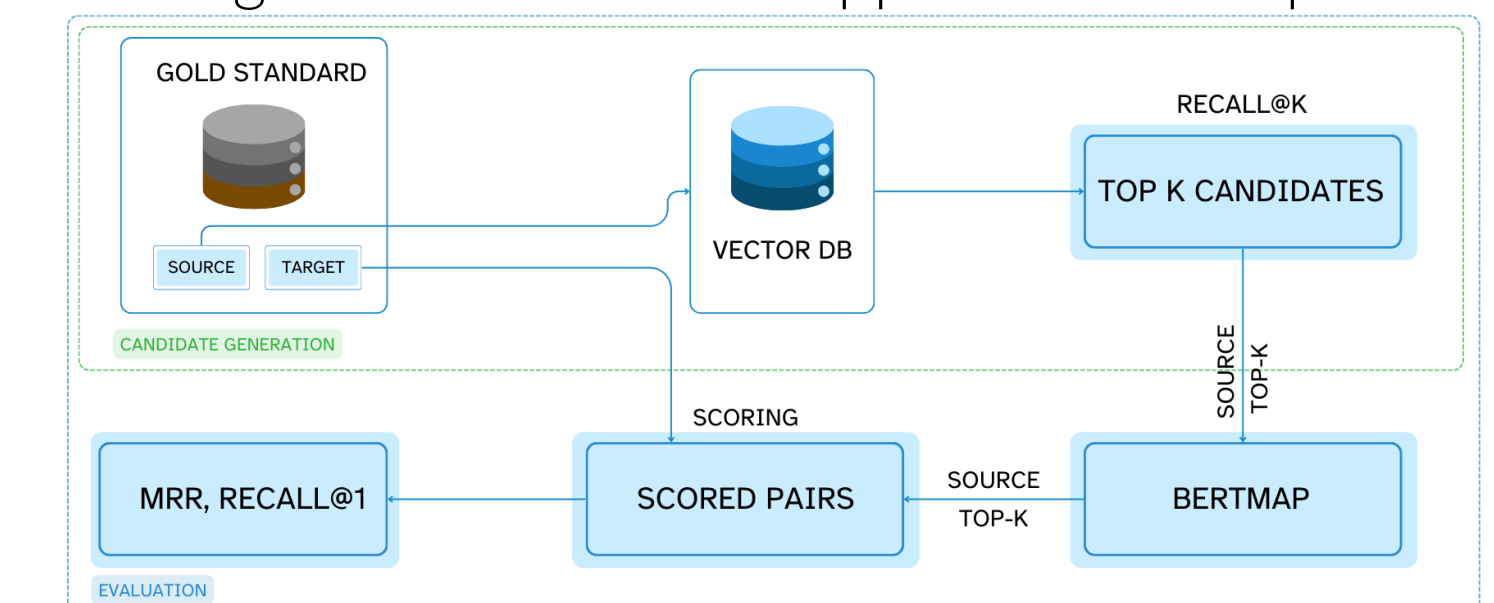


Figure 10. Evaluation pipeline

Candidate Retrieval Results (Recall@K)

Table 1. Candidate recall (Recall@K) under different setups and retrieval directions.

Model	Data	HP → MP			MP → HP		
		@50	@100	@200	@50	@100	@200
BERTMap	Unenr.	0.620	0.660	0.700	0.660	0.700	0.730
SapBERT	Unenr.	0.830	0.864	0.899	0.835	0.878	0.910
SapBERT	Enr.	0.827	0.861	0.893	0.834	0.864	0.902
SapBERT-FT	Unenr.	0.866	0.896	0.928	0.886	0.916	0.943
SapBERT-FT	Enr.	0.870	0.903	0.933	0.886	0.914	0.940

- SapBERT vector retrieval outperforms BERTMap.
- Enrichment is direction-sensitive

Results

Table 2. BERTMap scoring ablation (Recall@K and MRR @ 50) without Mapping and Repair

Retriever	Data	HP → MP			MP → HP		
		@1	@10	MRR	@1	@10	MRR
BERTMap	Unenr.	0.299	0.493	0.368	0.303	0.527	0.382
SapBERT	Unenr.	0.379	0.528	0.428	0.637	0.839	0.711
SapBERT	Enr.	0.422	0.559	0.466	0.567	0.809	0.652
SapBERT-FT	Unenr.	0.383	0.529	0.433	0.649	0.881	0.735
SapBERT-FT	Enr.	0.422	0.562	0.469	0.580	0.847	0.672

Conclusion

Stronger candidate retrieval raises alignment performance across ontologies, with SapBERT and SeMRA providing consistent gains without modifying BERTMap.